AGROBIOMETRIX: 1-YEAR DEPLOYMENT BUDGET & PAYMENT GUIDE

Annual Financial Model, Exact API Selections, Dollar-to-Naira Equivalents, and Immediate Activation Checklist

Target Horizon: 12 Full Months (2026–2027) | **Benchmark FX Rate:** ■1,650 / $1.00 USD | **Platform Scope:** All 8 Intelligence Suites | **Status:** Ready for Immediate Payment

1. Executive Summary & Investment Tiers

To deploy AgroBiometrix with **100% full functionality** for one year, the financial architecture is organized into two practical deployment tiers. Every recommended API and service has been specifically selected to guarantee zero downtime, native African tonal voice synthesis, high-speed multi-agent reasoning, and automated community radio syndication.

TIER 1: FULL PRODUCTION DEPLOYMENT
$600.00 USD / Year
■990,000 NGN / Year
Covers 24/7 dedicated cloud VM, custom domain, Gemini 1.5 Flash reasoning, YarnGPT/Google African TTS, podcast CDN, and contingency buffer.

TIER 2: ENTERPRISE + LAST-MILE GSM DISPATCH
$1,000.00 USD / Year
■1,650,000 NGN / Year
Includes everything in Tier 1 PLUS pre-funded Africa's Talking (AT) telecom airtime for automated 2G phone voice robocalls and WhatsApp broadcast audio notes.

2. Itemized Annual Budget Breakdown (Selected APIs & Infrastructure)

Component / Function	Selected Provider & Specific Tier	Billing Model	Monthly ($)	Annual ($USD)	Annual (NGN ■)
1. Cloud Server Hosting 24/7 2 vCPU, 4GB RAM, Linux Docker, zero spin-down	**DigitalOcean Basic Droplet** or **AWS Lightsail** (4GB RAM)	Fixed Monthly VM	$20.00	**$240.00**	■396,000
2. Custom Domain & SSL agrobiometrix.org with DNS & SSL	**Namecheap / Cloudflare Registrar** (.org domain + SSL)	Annual Renewal	$1.25	**$15.00**	■24,750
3. Core Multi-Agent AI Reasoning across all 8 suites	**Google Gemini 1.5 Flash API** ($0.075 / 1M input, $0.30 / 1M output)	Pay-as-you-go (~10k queries/mo)	$10.00	**$120.00**	■198,000
4. African Neural Voice AI Indigenous spoken audio (ha, yo, sw, fr)	**YarnGPT Pro API** + **Google Cloud TTS** (Chirp/Neural2)	Usage credits (~2,000 broadcasts/mo)	$15.00	**$180.00**	■297,000
5. Podcast Audio CDN MP3 enclosure hosting for FM stations	**Cloudflare R2 Storage** (10 GB free tier + zero egress fees)	Usage Storage	$1.25	**$15.00**	■24,750
6. Foreign FX & Buffer Buffer for card rate fluctuations (5%)	**Contingency Reserve**	One-time reserve	$2.50	**$30.00**	■49,500
SUBTOTAL: TIER 1 (PRODUCTION BASE)	**Complete 8-Suite Web & Broadcast Engine**	**12 Months**	**$50.00**	**$600.00**	**■990,000**
7. GSM Telecom Robocalls Push-to-phone 2G audio for rural farmers	**Africa's Talking (AT) Telecom API** (~■12/min voice call, ~■4/SMS)	Prepaid Airtime Wallet (~5,000 calls)	$20.80	**$250.00**	■412,500
8. WhatsApp Voice Delivery Voice note dispatch to extension agents	**Twilio WhatsApp Business API** ($0.005/msg outbound)	Prepaid Messaging Credits	$12.50	**$150.00**	■247,500
TOTAL: TIER 2 (FULL ENTERPRISE + TELEPHONY)	**End-to-End Lab to Farmer Ear Dissemination**	**12 Months**	**$83.30**	**$1,000.00**	**■1,650,000**

3. Immediate Payment & API Subscription Action Plan

To activate full functionality right away, follow this step-by-step subscription checklist:

Step 1: Google Gemini 1.5 Flash API ($120 / year)

- **Where to Subscribe:** Navigate to https://aistudio.google.com/ and sign in with your Google account.
- **How to Pay:** Click *'Get API Key'*, link a Google Cloud Billing account (accepts standard Visa/Mastercard, Chipper USD, or domiciliary cards).

- **Where to Paste in App:** Open AgroBiometrix, go to **Suite 7: Knowledge Network & API Hub** (or sidebar), and paste your key into Google Gemini API Key. It is immediately encrypted.

Step 2: YarnGPT / African Indigenous Voice AI ($180 / year)

- **Where to Subscribe:** Register on https://yarngpt.com/ or subscribe to Google Cloud Text-to-Speech at https://console.cloud.google.com/apis/library/texttospeech.googleapis.com.
- **How to Pay:** Select the Pro/Developer tier and fund with $15/month or an annual $180 upfront credit.
- **Where to Paste in App:** Navigate to **Suite 8: Last-Mile Vernacular Radio & Voice > Tab 3 (African Neural Voice Engine)** and enter your API key.

Step 3: Cloud Virtual Server & Custom Domain ($255 / year)

- **Domain:** Purchase agrobiometrix.org on Namecheap or Cloudflare Registrar ($15/year).
- **Server:** Deploy on DigitalOcean ($20/mo Basic Droplet, 4GB RAM) or AWS Lightsail (4GB RAM, $20/mo). Install Docker and run git clone && streamlit run app.py.

Step 4 (Optional for Tier 2): Africa's Talking Telecom Gateway ($250 / year)

- **Where to Subscribe:** Visit https://africastalking.com/ and register an account.
- **How to Pay:** Africa's Talking accepts direct local Nigerian bank transfer or Naira debit cards (Paystack/Flutterwave). Fund your wallet with ■412,500.

4. Recommended Payment Instruments for Nigerian Developers

Due to international spending limits on standard Naira debit cards ($20/month cap on domestic cards), use any of the following reliable channels to execute the USD subscriptions immediately:

- **Virtual USD Mastercard / Visa: Chipper Cash USD Card**, **Geegpay (Raenest)**, **Payday**, or **Grey.co**. These can be funded directly via instant Naira bank transfer and work seamlessly across Google Cloud, DigitalOcean, and Namecheap.
- **Domiciliary Bank Card:** GTBank, Access Bank, or Zenith Bank USD Domiciliary Debit Card.
- **Direct Local Naira Channels:** Africa's Talking (AT) natively bills in Nigerian Naira (NGN), so local GTB/Access/Zenith Naira cards work directly with zero exchange fees.

Budget Clearance: Approved for Operations	Annual Tier 1: $600 USD (■990,000 NGN)	Annual Tier 2: $1,000 USD (■1,650,000 NGN)